

What Technology Says About Decision-Making: Evidence From Cricket's Decision Review System (DRS)

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Ram Shivakumar¹

Abstract

We assemble and analyze a novel data set that provides evidence on the quality of decision-making by on-field umpires and teams in the sport of cricket. The decision review system, introduced in November 2009, gives teams a limited right to challenge an on-field umpire's decision. Of the 1,201 on-field umpire decisions that were challenged in test matches by nine playing teams between 2009 and 2014, 310 (25.81%) were reversed by a 3rd umpire. Controlling for several match and team characteristics, logistic regressions show that (1) leg before wicket (LBW) and caught decisions that are ruled "out" by the on-field umpire are *more* likely to be reversed than decisions that are ruled "not out"; (2) for both LBW and caught decisions, home and away teams are *equally* likely to win a 3rd umpire reversal.

Keywords

cricket, 3rd umpire, challenge, caught, DRS (decision review system), LBW, logistic regression, odds ratio, on-field umpire, reversed, upheld

Anyone watching a game of sport today on TV can, with the benefit of slow motion replays and other technological aids, immediately tell whether an on-field umpire's decision (OFUD) is incorrect. To prevent umpiring errors from influencing the

¹ Booth School of Business, University of Chicago, Chicago, IL, USA

Corresponding Author:

Ram Shivakumar, Booth School of Business, University of Chicago, 5807 S. Woodlawn Ave., Chicago, IL 60637, USA.

Email: ram.shivakumar@chicagobooth.edu

outcome of games, the administrators of sport have provided teams with a limited right to challenge an OFUD. Modern technology, that is, high-definition cameras, ball flight simulators, computer graphics, microphones, and so on, is used to reassess contentious plays and deliver final verdicts. For instance, in tennis, a player's challenge to an umpire's decision is resolved by the use of hawk-eye—a computer system that visually re-creates the path of the actual tennis ball to determine whether the ball is likely to have landed inside or outside the tennis court.

In November 2009, the International Cricket Council (ICC, 2009), which administers the game of cricket, introduced a new set of rules to adjudicate decisions that is known as the umpiring decision review system (DRS). The DRS provides teams with a maximum of two unsuccessful opportunities per inning¹ to challenge the decision of an on-field umpire. Responding to a challenge by one of the playing teams, a 3rd umpire, with the aid of multiple technologies that are described below, can uphold or reverse the decision of an on-field umpire. In the event that the 3rd umpire does not find evidence that contradicts the on-field umpire's call, the OFUD is upheld and the team (that challenged the OFUD) loses one opportunity to challenge. In the event that the 3rd umpire finds evidence that contradicts the on-field umpire's call, the OFUD is reversed and the team (that challenged the decision) retains its right to challenge an OFUD. If a team is on the wrong end of both of its challenges, it cannot challenge any more OFUDs during that inning.

The DRS and the technologies used to support it have provoked many passionate debates among cricket players (current and former), cricket writers, and fans (see, for instance, Chappell, 2013; Cozier, 2009). Proponents of the DRS view it as an improvement over the status quo. Some critics of the DRS have argued that the technologies used to support the DRS are flawed and are likely to distort the way the game is played (see Crowe, 2014). Some other critics of the DRS have argued that the DRS will ultimately lead to the loss of authority for, and trust in, on-field umpires (see Steen, 2011).

Despite the large number of opinion columns in newspapers, magazines, websites, and blogs on the DRS, what is surprising is that there has been no empirical assessment of the DRS and its influence on decision-making by umpires and teams. Cricket is a particularly good laboratory to study the link between the use of modern technologies and decision-making for at least two reasons. First, on-field umpires apply a great deal of subjective judgment in adjudicating plays. For instance, one of the most contentious forms of dismissal in cricket is the leg before wicket (LBW) decision, where the umpire is called upon to rule whether the ball has hit a batsman's pad and whether the ball would have gone on to hit the wicket.² On average, four batsmen are declared out LBW in every test match (see Sacheti, Patton, & Smith, 2015) though, as anyone who has watched a test match can attest to, the number of appeals (by the fielding team) for an LBW is generally several multiples of four. The DRS provides teams with an opportunity to receive restorative justice though, given the limited rights to challenge OFUDs, teams must use their challenges prudently.

Second, the on-field umpires and players have to make their decisions rapidly and are prohibited by ICC rules from consulting with others outside the playing field. For instance, an on-field umpire usually responds to an appeal by the fielding team within a matter of seconds. Furthermore, teams challenging the OFUD must do so within 30 s of the on-field umpire making his decision. Since test matches are played for up to 5 days in a row (30 h of play), on-field umpires must display a great deal of sustained concentration.

The aim of this article is to conduct an empirical assessment of the DRS. At the outset, it is important to emphasize that we do not question the efficacy of the technologies that the 3rd umpire employs to render his decision. We take the functions, attributes, and capabilities of the technologies used by the 3rd umpire to make his decision as a given. We provide answers to two questions that will be of interest to cricket (and sport) aficionados as well as academics. First, what does modern technology say about the accuracy of the decisions made by the on-field umpires? If on-field umpires are shown to make “few” errors, then ontological authority has been legitimized. If on-field umpires are shown to make “many” errors, an obvious implication is that on-field umpires have to be selected more carefully and that they must receive more training and guidance. Importantly, an assessment of the types of errors that on-field umpire errors make can help the ICC revise the guidelines and training systems required to reduce umpiring errors. Second, how is the 3rd umpire’s decision (to reverse or uphold the OFUD) related to the other variables that also exert an influence on the outcome? To be sure, the details surrounding each contentious play matter. These details include match information and team characteristics such as the identity of the batsman and the bowler, the nature of the challenge, the OFUD, the identity of the player/team that is challenging the decision, and so on.

We assemble and analyze a novel data set that encompasses the entire sample of DRS challenges and 3rd umpire decisions (3RDUDs) in test matches played between 2009 and 2014. Of the 1,201 on-field umpiring decisions that were challenged by the playing teams, 310 decisions (25.81%) were reversed by the 3rd umpire—a *conditional* on-field umpire error rate that is higher than the *unconditional* error rate (approximately 8%) attributed to umpires by the ICC. Controlling for the influence of several market and team characteristics, logistic regressions are used to assess the influence of several variables on the likelihood of the 3rd umpire reversing the OFUD.

The logistic regressions show that “out” decisions are 75% *more* likely to be reversed (to “not out”) as not out decisions (are to out) when the challenges involve LBWs, whereas out decisions are 111% *more* likely to be reversed (to not out) as not out decisions (to out) when the challenges involve catches, holding other variables constant. This result provides credence, though not proof, for the contention that with the advent of the DRS, on-field umpires appear to be less willing to give the benefit of the doubt to the batsman, a tradition, though not a rule, in cricket for more than a century (see, for instance, Findlay, 2011). The logistic regressions also show that for both LBW and caught challenges, one cannot reject the null hypothesis that

home and away teams are *equally* likely to win a 3rd umpire reversal. In other words, there is no evidence of bias (in favor of the either team) in the way that 3rd umpires adjudicate decisions.

Two recent articles examine decision-making by umpires in cricket. To test for the presence of bias in umpiring, Sacheti et al. (2015) study LBW decisions in 1,000 test matches played between 1986 and 2012. They find that when both on-field umpires are home umpires, the home team receives 16% fewer adverse decisions than visiting teams, although this home bias falls to 0.7% when both umpires are neutral umpires. They conclude that the source of the home team bias (in the early years of their sample) was outright favoritism rather than social pressure from home crowds. Given that there are two ways of improving decision-making in cricket—better technology (through higher speed cameras, for instance) or better umpires (through better training)—Borooah (2016) argues, using cost–benefit analysis, that the cost-efficient solution is to invest in more training for umpires.

It is worth noting, for the benefit of the reader not familiar with cricket and the DRS, that LBW and caught challenges have distinct standards for adjudication (by the 3rd umpire). With catches, the 3rd umpire is (usually) expected to find conclusive evidence as to whether the OFUD was correct or incorrect. With LBW challenges, hawk-eye produces a margin of error around its predicted ball path, so that the 3rd umpire is sometimes only able to conclude that the OFUD cannot be shown to be wrong. In such cases, the 3rd umpire chooses the “umpire’s call.”³

We mobilize our thoughts in four sections. The next section discusses the mechanics of the DRS and the technological aids that the 3rd umpire uses to make his decision. The following section discusses the data set that we have assembled and presents the descriptive statistical facts. In the penultimate section, we estimate logistic regressions that summarize the influence of match and team-specific variables on the 3rd umpire’s decision. A final section offers concluding remarks.

The DRS and Technological Aids

Figure 1 illustrates the mechanics of the DRS with the aid of a decision tree. At the first decision node, the on-field umpire responds to a query (an “appeal” in cricket jargon) from the fielding team by declaring the batsman out or not out. At the second decision node, the teams respond to the OFUD by challenging it or by not challenging it. If no challenge occurs, the OFUD is final and the game proceeds.⁴ If one of the teams challenges the OFUD, we proceed to the third decision node where the 3rd umpire, a qualified umpire who is sitting in a room outside the playing field that is equipped with TV monitors, gadgets, and other support systems (see below), reexamines the contentious play. If a review of the contentious play is unable to conclusively reject the OFUD, the OFUD is upheld. If a review of the play provides conclusive evidence that the OFUD is incorrect, the OFUD is reversed. In the event

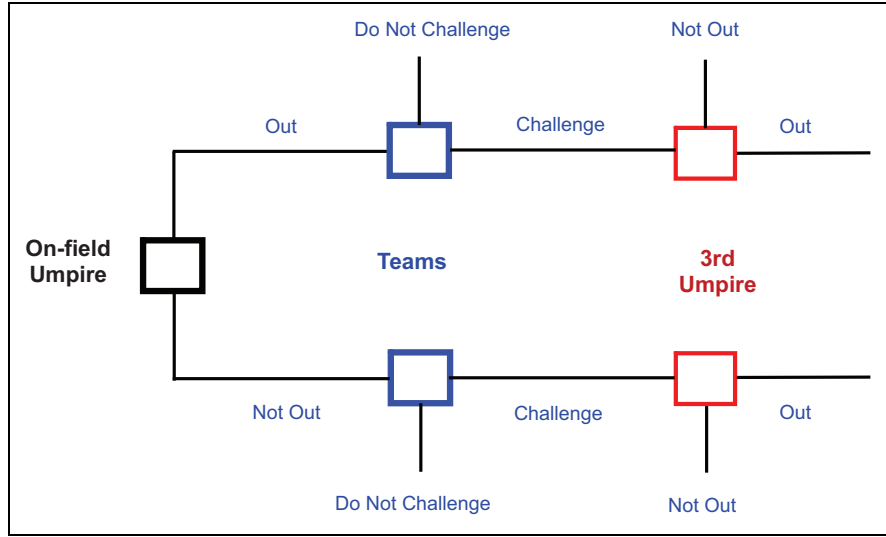


Figure 1. Decision tree.

of a reversal, the player who was declared out (not out) by the on-field umpire gets to continue (not continue).

The DRS is generally supported by three distinct technological aids: (1) hawk-eye, (2) hot spot, and (3) snickometer.⁵ Hawk-eye, a ball-tracking system which is also used in tennis and soccer, is essential in helping the 3rd umpire adjudicate LBW challenges. It relies on several high-definition cameras that record each play from multiple angles from the moment that the ball leaves the bowler's hand till. The cameras record the ball's speed, path, and bounce and upload the data frame by frame to a computer software system. This software system uses the data to predict the future path of the ball beyond the point where it made contact with the batsman, hence determining whether the ball would have gone on to hit the stumps. The still photograph on the top left-hand side of Figure 2 is an illustration of what the 3rd umpire sees when he watches a slow-motion replay of the contentious decision (image retrieved from <http://www.cricket.com.au/news/decision-review-system-up-for-debate-during-australia-india-test-series/2014-12-11>).

Hot spot is an infrared imaging system that helps determine whether the ball has made contact with the batsman's bat (or glove) or whether the ball has made contact with the batsman's pad. It is a vital technological aid in adjudicating both LBW and caught decisions.⁶ Hot spot is implemented with the use of infrared cameras that are positioned at opposite ends of the cricket ground. The cameras measure the heat friction generated by contact between the ball and the bat (or the pad) and generate a series of black-and-white frames that zero in on the point of contact. The still photograph on the top right-hand side of Figure 2 is an example of what the 3rd



Figure 2. Technologies used to support the decision review system.

umpire sees when he watches a slow-motion replay of the contentious decision (image retrieved from <http://www.ebuyer.com/blog/2015/02/tech-in-cricket/>).

The snickometer, a technology used to assess challenges involving catches, provides audiovisual evidence on whether the ball might have made contact with the batsman's bat. In the event that the ball has made contact with the batsman's bat, an elevated sound wave will appear on the screen at the moment when the ball passes the bat. The still photograph at the bottom of Figure 2 is an example of what the 3rd umpire sees when he reviews a catch challenge (image retrieved from <http://www.ebuyer.com/blog/2015/02/tech-in-cricket/>).

Data and Descriptive Statistics

The data used in this study are assembled from the text commentary of test matches (provided in real time) from ESPN Cricinfo's website (www.cricinfo.com). We catalog every DRS challenge in test matches from the introduction of the system in November 2009 through December 2014.⁷ In addition, we catalog other information such as the index number and dates of the test match, the identity of the home and away teams, the city (specifically, the cricket ground) in which the match was played, the identity of the team that initiated the challenge, the identity of the batsman and bowler when the on-field umpiring decision was challenged, the batsman's rank (whether top 7 or bottom 4 as listed on the team sheet), the bowler's style (fast or spin), the type of challenge (LBW or caught), the identity of the on-field umpire, the stage of the game when the challenge was initiated (over #), the decision that the on-field umpire made, and importantly, the decision of the 3rd umpire.

Table 1 provides a summary of the data set. It shows that there were 1,201 challenges by the playing teams during the 126 test matches in which the DRS was employed. Among the many statistical facts that will be of interest to cricket aficionados and others, two stand out. First, 310 of the 1,201 challenges were reversed by the 3rd umpire—a reversal rate of 25.81%. In simple language, more than one in every four decisions made by the on-field umpire and challenged by one of the playing teams is reversed. The ICC claims that umpires get more than 91% of their decisions right (see Steen, 2011). However, they are not explicit in telling us what their sample includes and excludes. Presumably their sample does not count the many on-field decisions (out and not out) that are patently obvious and do not require the on-field umpire to render a judgment. The ICC also does not disclose how it separates the many frivolous appeals that fielding teams make (in optimism rather than conviction) from the more serious appeals (that are not challenged after due contemplation). Second, teams (as a group) challenge approximately 10% *more* decisions in away games than in home games. This *may* be because (a) teams have greater familiarity with playing conditions at home and less familiarity with playing conditions away from home and/or (b) home teams still benefit (relative to visiting teams) from on-field umpire bias (despite the presence of neutral umpires), so teams

Table 1. Decision Review System Use by Teams.

Team	Matches	Challenges	Reversals	Reversal Rate (%)	Home Challenges	Reversals	Away Challenges	Reversals
Australia	45	197	58	29.44	92	30	105	28
England	45	187	55	29.41	92	30	95	25
New Zealand	30	131	34	25.95	63	18	68	16
South Africa	34	154	42	27.27	82	22	72	20
Pakistan	30	194	42	21.65	90	13	104	29
Sri Lanka	26	148	35	23.65	67	14	81	21
West Indies	30	127	28	22.05	70	12	57	16
Zimbabwe	6	35	11	31.43	0	0	35	11
Bangladesh	6	28	5	17.86	11	1	17	4
Total	126	1,201	310	25.81	567	140	634	170

		3 RD UMPIRE'S DECISION	
		OUT	NOT OUT
ON-FIELD UMPIRE'S DECISION	OUT	351	159
	NOT OUT	151	540

Figure 3. Decisions by on-field umpires and 3rd umpires.

have to challenge more OFUDs in away games to offset the advantage that accrues to home teams.

Figure 3 is a summary matrix of all OFUDs that were challenged by the playing teams. The vertical axis displays the decision of the on-field umpire—out or not out—while the horizontal axis identifies the decision of the 3rd umpire—out or not out. The diagonal terms are those decisions that the on-field umpire got correct while the off-diagonal terms are those decisions that the on-field umpire got wrong. The top row of the matrix measures the challenges made by the batting team and the bottom row of the matrix measures the challenges made by the fielding team. Notice that fielding teams account for approximately 57% of all challenges though batting teams win a higher proportion of their challenges.

Table 2 provides a breakdown of DRS decisions by type of challenge: “LBW” and “caught.” Notice that LBW challenges constitute more than 75% of all challenges though 3rd umpire reversals are less frequent for LBW challenges than caught challenges. Readers will also notice the greater dispersion in reversal rates across countries for caught decisions as opposed to LBW decisions.

Figure 4(a) and (b) provides a more granular look at the LBW and caught challenges. They identify the combination of out and not out decisions taken by the on-field and 3rd umpires, respectively. From Figure 4A, it is easy to show that the odds of an LBW being reversed from out to not out is equal to the probability of an out being reversed to not out divided by the probability of an out being upheld as out or $(119/420)/(301/420) = 0.2833/0.7166 = 0.3953$. The odds of an LBW being

Table 2. Breakdown of Decisions by Type of Challenge.

Teams	LBW Challenges	Reversals	Reversal Rate (%)	Caught Challenges	Reversals	Reversal Rate (%)
Australia	140	37	26.43	57	21	36.84
England	134	32	23.88	53	23	43.40
New Zealand	106	27	25.47	25	7	28.00
South Africa	107	25	23.36	47	17	36.17
Pakistan	159	32	20.13	35	10	28.57
Sri Lanka	113	28	24.78	35	7	20.00
West Indies	105	19	18.10	22	9	40.91
Zimbabwe	28	6	21.43	7	5	71.43
Bangladesh	20	4	20.00	8	1	12.50
Total	912	210	23.03	289	100	34.60

reversed from not out to out is equal to the probability of a not out being reversed to out divided by the probability of a not out being upheld as not out or $(91/492)/(401/492) = .1849/.8150 = .2260$. Hence, the odds ratio (*OR*) of an LBW challenge that is given out by the on-field umpire being reversed (relative to an LBW challenge that is given not out being reversed) is equal to 1.7514. That is, LBW challenges that are given out are about 75% *more* likely to be reversed (than LBW challenges that are given not out). The logistic regressions in the next section provide further evidence on the robustness of this result.

From Figure 4B, it is possible to show that the odds of a caught being reversed from out to not out is equal to the probability of an out being reversed to not out divided by the probability of an out being upheld as out or $(40/90)/(50/90) = 0.4444/0.5555 = 0.80$. The odds of a caught being reversed from not out to out is equal to the probability of a not out being reversed to out divided by the probability of a not out being upheld as not out or $(60/199)/(139/199) = 0.3015/0.6984 = 0.4317$. Hence, the *OR* of a caught challenge that is given out by the on-field umpire being reversed (relative to a caught challenge that is given not out being reversed) is equal to $0.80/0.4317 = 1.8531$. That is, caught challenges that are given out are about 85% *more* likely to be reversed (than caught challenges that are given not out). The logistic regressions in the next section provide further evidence on the robustness of this result.

Logistic Regressions

We estimate logistic regressions of the form:

$\text{Log}[q_i/(1 - q_i)] = b_0 + b_1X_{1,i} + b_2X_{2,i} + b_3X_{3,i} + \dots + \varepsilon_i$, where $q_i = p(Y_i = 1)$ is the probability of a reversal by the 3rd umpire (Y_i is the i th observation of the

LBW		Caught	
3 RD UMPIRE'S DECISION		3 RD UMPIRE'S DECISION	
OUT	NOT OUT	OUT	NOT OUT
OUT ON-FIELD UMPIRE'S DECISION	301	50	40
	91	60	139
NOT OUT		NOT OUT	
ON-FIELD UMPIRE'S DECISION		ON-FIELD UMPIRE'S DECISION	
OUT		OUT	
119		401	

Figure 4. Further breakdown of decision review system use by type of challenge.

Table 3. Regression Variables.

Dependent variable	Explanation
3rd Umpire decision	Likelihood that 3rd umpire reverses (1) or upholds (0) on-field umpire's decision
Independent variables	
On-field umpire decision	Whether on-field umpire gives batsman out (1) or not out (0)
Challenge team characteristic	Whether the challenging team is home team (1) or away team (0)
Control variables	
Challenge team ID	Dummy variables (1 or 0) for 8 countries relative to Australia
Umpire ID	Dummy variables (1 or 0) for 24 umpires relative to K. Dharmasena, the reference umpire
Location	Dummy variables for match location (1 or 0) relative to Australia, the reference country
Bowler characteristic	Whether bowler is a fast (1) or slow (0) bowler
Batsman characteristic	Whether batsman is a top seven batsmen (1) or bottom four batsmen (0)
Challenge team rank	Challenging team's International Cricket Council rank (1 through 9) at the time the match was played
Time	Values from 1 through 6 for each of the years in which the match was played

dependent variable), the X s are the predictor variables, and the β s are the estimated regression coefficients.

Table 3 is a list of the variables used in the logistic regressions. The dependent variable, 3RDUD, is a binary variable that takes on a score of 1 when the 3rd umpire *reverses* the OFUD and a score of 0 when the 3rd umpire *upholds* the OFUD. The first independent variable, OFUD, takes on a value of 1 when the OFUD is out and a value of 0 when the OFUD is not out. The *OR* for OFUD provides an estimate of the likelihood of a 3rd umpire reversal when the on-field umpire gives a decision of out as opposed to a decision of not out, holding other variables fixed. The second independent variable, *challenge team characteristic* (CTC) takes on a value of 1 when the team challenging the OFUD is the home team and a value of 0 when the team challenging the OFUD is the away team. The *OR* for CTC will tell us whether home teams are more (or less) likely to win reversals than away teams, holding other variables constant.

Six covariates are included in all the regressions. The variable, *batsman characteristic*, takes on a value of 1 when the batsman involved in the challenge is a top-order batsman (listed in the team sheet as a top seven batsman) and a value of 0 when the batsman involved in the challenge is a lower order batsman (listed in the team sheet as a bottom four batsman). The variable, *bowler characteristic*, takes on a

Table 4. Logistic Regressions.

Leg before wickets (LBWs)		Catches	
Variable	Exp β (<i>p</i> value)	Variable	Exp β (<i>p</i> value)
OFUD LBW	1.752 (.001)	OFUD CAUGHT	2.115 (.003)
CTC LBW	1.070 (.683)	CTC CAUGHT	0.932 (.810)
Constant	2.198 (.130)	Constant	0.246 (.124)
Controls	Yes	Controls	Yes
No. of observations	912	No. of observations	289
Nagelkerke s^2	0.062	Nagelkerke R^2	0.187
Hosmer–Lemeshow (significance)	0.987	Hosmer–Lemeshow (significance)	0.476

Note. CTC = challenge team characteristic; OFUD = on-field umpire decision.

value of 1 when the bowler involved in the challenge is a fast bowler and a value of 0 when the bowler involved in the challenge is a spin bowler. The variable, *umpire id*, is a dummy variable that takes on a value of 1 when the on-field umpire involved in the challenge is, for instance, Aleem Dar, and 0 otherwise. The model contains 17 dummy variables, one for each umpire, with K. Dharmasena as the reference umpire.⁸ The variable, *challenge team id*, is a dummy variable that takes on a value of 1 when the challenging country is a given country (say England) and 0 otherwise. Since we have nine countries in the sample, the model contains eight dummy variables with Australia as the reference country. The variable, *challenge team rank*, takes on an integer value between 1 and 9 (reflecting the ICC's rank ordering of teams in the test match format at the time that the test match was played). The variable, *location*, is a dummy variable that takes on a value of 1 if the match is held in a given country (say England) and 0 otherwise. Since test matches were held in eight countries during the sample period,⁹ the model contains seven dummy variables with Australia as the reference country.

Table 4 reports the results from a pair of logistic regressions—the first for LBW challenges and the second for caught challenges. For each regression, we report the estimated *OR* (Exp β) for each variable as well as the corresponding *p* value (in parentheses). We also report the estimated Nagelkerke R^2 , a pseudo coefficient of determination that takes on a value between 0 and 1, and the estimate from a Hosmer–Lemeshow test—a χ^2 goodness-of-fit test.

From inspection (of the *p* values) of the pair of regressions, notice that OFUD|LBW and OFUD|CAUGHT are statistically significant. The *OR* of 1.752 for OFUD|LBW tells us that out decisions are 75.2% *more* likely to be reversed from out to not out than vice versa, while the *OR* of 2.115 for OFUD|CAUGHT tells us that out decisions are about 111.5% *more* likely to be reversed from out to not out than vice versa.

To understand the importance of these results, it is worth noting that for more than a century, one of the unwritten rules of cricket has been that on-field umpires give the benefit of doubt to batsman. That is, if umpires believe that there is even a small probability that batsmen (involved in a contentious play) are not out, they are to be given not out.¹⁰ Some cricket players and journalists have observed that the advent of the DRS has upset this tradition because on-field umpires appear to be willing to give batsman out more frequently than earlier (see Dobell, 2012; Steen, 2011).

As Peter Willey (2009; the former English umpire) notes:

I do find myself thinking differently about decisions, especially, LBWs. And I shouldn't think like that. I have been umpiring for 15 years and for the last 13 certain decisions were definitely not out- now due to television I think they are close . . . when the technology is used fully a lot more people are given out. Reassessing my decision-making means I am giving decisions based on Hawk-Eye graphics on the television.

Findlay (2011) bluntly summarizes the impact of the DRS on on-field umpiring decisions pertaining to LBWs:

The DRS has clearly shown that umpires have been too conservative in considering lbw appeals in the past, and that batsmen have been getting away with murder for years! The higher incidence of lbw dismissals for both spin and quick bowlers is a result not only of the direct intervention of video replays, but also, in all probability, of a realisation by umpires that they are safer in giving out what they originally would have considered to be marginal decisions only a year or two ago.

What will likely surprise many followers of cricket is that the greater likelihood of "outs" being reversed to not out extends to caught decisions as well.

While there is some evidence (in cricket and other sports) that home teams receive the marginal calls in their favor (see Moskowitz & Wertheim, 2012), notice (from inspection of the p values in Table 4) that the variable, CTC, is statistically insignificant in both sets of regressions. Consequently, one cannot reject the null hypothesis that home and away challenges are *equally* likely to win (or lose) reversals from the 3rd umpire. In simple language, we find no evidence that home teams have any advantage over visiting teams in winning 3rd umpire reversals.

Conclusion

Technology is exerting a significant influence on the way sport is adjudicated and played. However, with the exception of anecdotal evidence, precious little is known about the quality of decision-making by on-field umpires. We take a first step toward redressing this imbalance by assembling and analyzing a novel data set that provides evidence on the quality of decision-making by on-field umpires in test match cricket.

Our analysis leads to three conclusions. First, more than one in every four decisions made by the on-field umpire is incorrect. Second, for both LBW and caught challenges, OFUDs of out are *more* likely to be reversed by the 3rd umpire to not out than OFUDs of not out. Third, home and away teams are *equally* likely to have their challenges successfully reversed by the 3rd umpire.

These results have important implications for way cricket is played, adjudicated by on-field and 3rd umpires and administered by the ICC. First, since teams have a limited right to challenge OFUDs and since technology is getting better at exposing umpiring errors, the pressure on on-field umpires to reduce errors, and on the ICC to amend the DRS, will likely increase. Cricket is relatively unique among team sports (like American football, soccer, and baseball) that have instituted a review of on-field umpiring decisions in three significant ways: (a) cricket uses technology more intensively than other sport to resolve differences of opinion over umpiring decisions. This is attributable, in part, to the fact that test match cricket is, unlike other games of sport, played over a 5-day period with the ultimate result (win, lose, or draw) often depending on a small number of decisions by the on-field umpires. (b) The players on the cricket field, rather than coaches who have access to greater information, are the ones who must choose in real time whether to challenge an OFUD. In major league baseball (MLB), the managers of the respective teams, the only ones with the authority to challenge umpiring decisions, will have likely received a green light from their assistants before they invoke their authority to challenge. (c) In cricket, on-field umpiring errors are generally corrected only when the playing teams use their limited right to challenge decisions. The English Premier League (EPL) has begun to use goal-line technology, a system that instantaneously alerts the referee and match officials by transmitting a signal to the referee's watch. In theory, there is no limit to the number of times the referee can correct his own decision.¹¹

Second, on one important dimension, the DRS can be said to be working quite well: There is no evidence of bias against home or away teams in the way that the 3rd umpires make their decisions. Given the paucity of umpires on the elite panel (umpires deemed to be the best), it has been suggested by some writers that at least one home umpire should be allowed to umpire in test matches (see Stewart, 2013). However, given the limited number of challenges that each team can call upon, and most importantly, our limited knowledge of the quality of on-field umpiring decisions when OFUDs are not challenged, we recommend that cricket administrators proceed with caution.

The analysis conducted here ought to be extended in several ways. First, relatively little is known about how the quality of decision-making varies across teams and across umpires, whether this variation is significant and the nature of the policies and systems that might be adopted to reduce such variance. Second, relatively little is known about the quality of on-field umpiring when teams do not challenge on-field umpire decisions. The analysis of these questions will provide a more comprehensive picture of the quality of decision-making by on-field umpires and teams.

Author's Note

Errors of commission and omission are mine alone.

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Notes

1. When the decision review system (DRS) was introduced in 2009, teams were given a maximum of two unsuccessful challenges per innings. In 2013, the International Cricket Council modified this rule to allow teams an additional maximum of two (unsuccessful) challenges every 80 overs of an inning.
2. For a batsman to be given out leg before wicket (LBW), the laws of the game state that the ball (1) did not pitch outside the leg stump, (2) hit the batsman's pad first (before contact with the bat), (3) hit the batsman in line with the wicket, and (4) would have gone on to hit the wicket. If the umpire doubts that any one of the four conditions identified above are not satisfied, he is obliged to give the batsman "not out."
3. The Cricinfo text commentary does not identify 3rd umpire decisions that were adjudicated as "umpire's call."
4. Note that our data set only catalogs on-field umpire decisions that are challenged by the playing teams.
5. In many test matches in our sample, hawk-eye was the only technology used to aid the 3rd umpire.
6. For LBW challenges, hot spot ascertains whether the ball made contact with the batsman's pad before hitting anything else while for caught challenges or whether the ball made contact with the bat before being caught by a fielder.
7. The data set begins with test # 1934 between New Zealand and Pakistan and runs through test # 2150 between South Africa and New Zealand. Test matches in which the DRS was not in use are excluded from the analysis.
8. Since 6 (of the 24) umpires in the data set were involved in less than 12 challenges, we drop them from the model.
9. There were no test matches held in Zimbabwe in which the DRS was in use.

10. One piece of evidence that bowlers (and fielding teams) found it very difficult to win an LBW from the on-field umpire (in the pre-DRS era) is that spin bowlers won a total of four L decisions in the 1975, 1979, and 1983 World Cups (when the matches were played in England). In the 2011 World Cup (played in India), the proportion of LBWs given by the on-field umpire (and upheld by the 3rd umpire) rose to 16.28% of all decisions confirmed against the batsman (see Findlay, 2011).
11. In soccer, a goal is considered to be scored when the whole ball crosses the line between the goal posts and the crossbar. In most cases, it is patently clear to everyone whether the ball has crossed the goal line. However, in a small number of cases, it is difficult for the referee to tell whether a ball has crossed this line before it was saved by the goal keeper or cleared by a defender.

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Author Biography

Ram Shivakumar is a professor of economics and strategy at the Booth School of Business and a managing director at the Berkeley Research Group.